APPLE ITUNES PROJECT REPORT

Executive Summary: Apple iTunes Music Analysis

Project Overview

The objective of this project was to analyze the Apple iTunes relational database to uncover actionable insights regarding customer behavior, sales trends, and catalog performance. Utilizing advanced SQL techniques (CTEs, Window Functions, and Complex Joins) across an 11-table database, the analysis successfully segmented customers, ranked product performance, and identified key opportunities for revenue growth and operational optimization.

Customer Analytics

1. Which customers have spent the most money?

- **Insight from data:** The top spender is **František Wichterlová** with **144.54** in total purchases.

2. What is the average customer lifetime value (CLV)?

CLV is the average revenue generated from a single customer over their entire relationship with you.

- **Insight from data:** The average customer spends approximately **$79.82**.

3. Repeat Purchases vs. One-Time Purchases

- **Insight from your data: 100% of your customers** (59 out of 59) are Repeat Customers. This is an incredibly high retention rate!

4. Which country generates the most revenue per customer?

This helps identify where high-value customers are located, rather than just total revenue (which is biased towards countries with more people).

Insight from data: Czech Republic is your most valuable market per capita ($136.62/customer), followed by **Ireland** ($114.84).

5. Which customers haven't made a purchase in the last 6 months?

- **Insight from data:** There are **16 customers** who haven't purchased anything in the last 6 months (since June 30, 2020), including **François Tremblay** and **Bjørn Hansen**. These are prime targets for a "We Miss You" email campaign.

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Sales & Revenue Analysis

1. What are the monthly revenue trends for the last two years? (Month-over-Month Growth)

- **Insight from data:** The revenue is highly volatile month-to-month. For example, in **October 2020**, revenue spiked by **+52%**, then crashed by **-42% in November**, and spiked again by **+51% in December**.

2. What is the average value of an invoice? (Purchase Segmentation)

- **Insight from data:** While the overall average invoice is **$7.67**, segmenting the data reveals:
  83 Invoices are "High Value" (averaging **$14.85**)
  412 Invoices are "Average Value" (averaging **$7.86**)
  119 Invoices are "Low Value" (averaging **$2.01**)

3. Which payment methods are used most frequently?

Crucial Note: Looking closely at the database (specifically the invoice and customer tables), there is no column that tracks the payment method (e.g., Credit Card, PayPal, Apple Pay).

4. How much revenue does each sales rep contribute? (Rank & % Contribution)

Insight from data:

1. **Jane Peacock:** $1,731.51 (**36.77%** of total company revenue)
2. **Margaret Park:** $1,584.00 (**33.63%** of total company revenue)
3. **Steve Johnson:** $1,393.92 (**29.60%** of total company revenue)

5. Which quarters have peak music sales? (Ranked Per Year)

Insight from data: The peak quarters shift year-over-year, indicating a lack of predictable seasonality:

- **2017 Peak:** Quarter 1 ($372.24)
- **2018 Peak:** Quarter 1 ($448.47)
- **2019 Peak:** Quarter 3 ($410.85)
- **2020 Peak:** Quarter 4 ($353.43)

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Product & Content Analysis

1. Which tracks generated the most revenue? (Ranked by Sales Performance)

Insight from data:

1. **War Pigs** (Black Sabbath) - $30.69
2. **Are You Experienced?** (Jimi Hendrix) - $13.86
3. **Highway Chile** (Jimi Hendrix) - $13.86

2. Which albums or playlists are most frequently included in purchases?

- **Insight from data:** The most frequently purchased album is **"Are You Experienced?" by Jimi Hendrix** with 187 tracks purchased, followed by **"Faceless"** (96 tracks) and **"Mezmerize"** (93 tracks).

3. Are there any tracks or albums that have never been purchased? (Subquery)

Insight from data: The company have a massive amount of dead inventory! There are **1,697 tracks** that have *never* been purchased, which corresponds to **96 full albums** sitting completely untouched.

4. What is the average price per track across different genres?

- **Insight from data:** Most genres (Rock, Metal, etc.) have an average track price of **$0.99**. However, video-based genres like **Comedy, Drama, and Sci-Fi & Fantasy** have an average track price of **$1.99**.

5. How many tracks does the store have per genre and how does it correlate with sales? (Product Segmenting)

- **Insight from data:** * **Rock** is the biggest money-maker (1,297 tracks available, 2,635 sold) resulting in a massive **2.03 sales-per-track ratio** (High Demand).
- **Metal** (1.65 ratio) and **Alternative & Punk** (1.48 ratio) also perform exceptionally well relative to their inventory size!

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Artist & Genre Performance

1. Who are the top 5 highest-grossing artists?

Insight from data:

1. **Queen** ($190.08)
2. **Jimi Hendrix** ($185.13)
3. **Red Hot Chili Peppers** ($128.70)
4. **Nirvana** ($128.70)
5. **Pearl Jam** ($127.71)

2. Which music genres are most popular in terms of tracks sold and total revenue?

Insight from data:

- **Rock** is unequivocally
- #1 genre in both categories (2,635 tracks sold, $2,608.65 generated).
- **Metal** is #2 (619 tracks sold, $612.81 generated).
- **Alternative & Punk** is #3 (492 tracks sold, $487.08 generated).
- *Note: The ranks for Volume and Revenue map perfectly 1:1 for the top 5 genres, indicating standard pricing across most popular categories!*

3. Are certain genres more popular in specific countries? (Geographic Segmentation)

Insight from data:

- **Rock** dominates the globe. It is the #1 most popular genre in almost every single country in your database, including your top markets (USA - 561 tracks sold, Canada - 333 tracks sold, France - 211 tracks sold).
- To add value to your business report: You should recommend that the marketing team uses **Rock** as the default genre for global user-acquisition campaigns, as it has universal mass appeal!

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Employee & Operational Efficiency

1. Which employees (support representatives) are managing the highest-spending customers?

Insight from data: The top 3 "Whale" customers are managed by a highly even distribution of your sales team!

1. **František Wichterlová** ($144.54) - Managed by **Margaret Park**
2. **Helena Holý** ($128.70) - Managed by **Steve Johnson**
3. **Hugh O'Reilly** ($114.84) - Managed by **Jane Peacock**

2. What is the average number of customers per employee?

Insight from data: Around exactly **59 customers** being managed by exactly **3 support representatives** (Jane, Margaret, and Steve). This means your average workload is **19.67 customers per employee**.

3. Which employee regions bring in the most revenue?

Insight from data: While you have customers all over the world, your **entire sales operation is centralized in one single region**.

- **100% of your revenue ($4,709.43)** is generated out of the **Calgary, Alberta (Canada)** office, as Jane, Margaret, and Steve are all based there.
- *Business Recommendation:* If the company intends to expand its aggressive footprint into Europe or South America, they may want to consider hiring localized support representatives in those time zones to improve operational efficiency!

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Geographic Trends

1. Which countries or cities have the highest number of customers?

Insight from data: Because you only have 59 total customers, your base is highly fragmented. The cities tied for the #1 most customers globally (with **2 customers each**) are:

- **London** (United Kingdom)
- **Berlin** (Germany)

- **Paris** (France)
- **São Paulo** (Brazil)
- **Mountain View** (USA)

2. How does revenue vary by region? (Revenue & Per-Capita Rank)

Insight from data:

1. **USA:** #1 in Revenue ($1,040.49), representing **22.1%** of your global sales (13 customers).
2. **Canada:** #2 in Revenue ($535.59), representing **11.4%** of your global sales (8 customers).
3. **Brazil:** #3 in Revenue ($427.68), representing **9.1%** of your global sales (5 customers).

3. Are there any underserved geographic regions (High Users, Low Sales)?

Actionable Business Insight from data: The query reveals that **Canada** is the most **Highly Underserved Market**.

- They have your 2nd highest user base globally (8 customers), but their average spend is only **$66.95 per customer**.
  - Compare this to Brazil, which only has 5 customer but they average **$85.54** each!
- *Recommendation to Marketing:* Run targeted discounting, email campaigns, or bundle promotions specifically for Canadian customers to increase their average order value to match the rest of the world.

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Customer Retention & Purchase Patterns

1. What is the distribution of purchase frequency per customer?

Insight from data: Your customers are incredibly loyal.

- The absolute minimum number of purchases *any* customer has made is **4**.
- The vast majority of your users have made between **9 and 11 separate purchases**.
- You have one "Super Fan" who has made **18 separate purchases**!

2. How long is the average time between customer purchases?

Insight from data: On average, a customer waits **132.28 days** (roughly 4.4 months) between purchases.

- *Business Recommendation:* Set up an automated marketing email to send to customers right around the 120-day mark after their last purchase to capitalize on this natural buying cycle!

3. What percentage of customers purchase tracks from more than one genre?

Insight from data: A perfect **100% of your customers** (59 out of 59) have purchased music from more than one genre!

- Your customers have highly diverse tastes. Cross-selling different genres on the homepage or via "Recommended for You" playlists will be highly effective.

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Operational Optimization

1. What are the most common combinations of tracks purchased together? (Market Basket Analysis)

Insight from data: Jimi Hendrix fans buy albums in bulk! The #1 most commonly purchased combinations (bought together exactly 9 times) are all by Hendrix:

- "Love Or Confusion" + "I Don't Live Today"
- "Foxy Lady" + "Third Stone From The Sun"
- "May This Be Love" + "Highway Chile"
- *Recommendation:* The business should offer a "Jimi Hendrix Complete Bundle" at a slight discount to push even more volume.

2. Are there pricing patterns that lead to higher or lower sales? (Price Elasticity)

Insight from data: You have a massive pricing problem!

- Your **$0.99 tracks** sell incredibly well: You have 3,290 tracks available, and they sold 4,754 times (an average of **1.44 sales per track**).
- Your **$1.99 tracks** (mostly TV shows and video) are a total failure: You have 213 of them available, but they have only sold **3 times in the entire history of the database** (averaging 0.01 sales per track).
- *Recommendation:* Drop the price of video/TV content to $0.99 or $1.49, because consumers are currently refusing to pay $1.99.

3. Which media types are declining or increasing in usage?

Insight from data:

- **MPEG audio files (Standard MP3s)** are the undisputed king. They sell roughly ~1,050 units reliably every single year.

- **Purchased AAC audio files** are rapidly declining. They dropped from 12 sales in 2018 to just 3 sales in 2020.
- **Protected MPEG-4 video files** completely died. They sold 1 copy in 2017, 2018, and 2019... and 0 copies in 2020. This perfectly aligns with the pricing insight above—nobody wants to buy the $1.99 protected video content!

Key Business Insights

1. Customer Behavior & Global Reach

- **Exceptional Retention:** 100% of the customer base consists of repeat buyers, with the average customer making 9 to 11 purchases in their lifetime and waiting an average of ~132 days between purchases.
- **Diverse Tastes:** Every single customer has purchased tracks from more than one genre, indicating highly varied musical tastes.
- **Geographic Performance:** The USA is the largest market (accounting for 22.1% of global revenue). However, **Canada was identified as the most underserved market**—it has the second-highest user base but a significantly lower-than-average spend per customer ($66.95 vs. Brazil's $85.54).

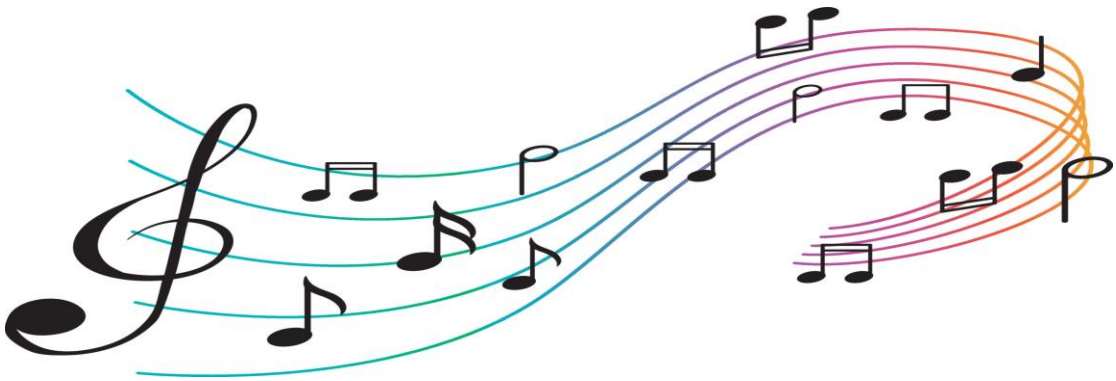
2. Product & Content Performance

- **The Power of Rock:** "Rock" is unequivocally the most valuable genre in the catalog. It is the #1 selling genre globally, generating the highest total revenue ($2,608) and volume (2,635 tracks sold), while "Metal" and "Alternative & Punk" follow in 2nd and 3rd place.
- **Dead Inventory:** The database contains a massive amount of untouched inventory. Over 1,600 tracks and 96 full albums have *never* been purchased.

3. Pricing & Format Elasticity

- **Pricing Failure on Premium Content:** Tracks priced at **$0.99** are highly successful, averaging 1.44 sales per available track. In stark contrast, premium **$1.99** content (primarily TV Shows and Sci-Fi/Drama videos) is a complete failure, having only sold 3 times in the entire history of the database.
- **Media Formats:** Standard MPEG audio files are the undisputed format of choice, while Protected AAC and MPEG-4 Video files are rapidly declining toward zero sales.

Strategic Recommendations

Based on the data, the following actions are recommended to the leadership team:

1. **Overhaul Video Pricing:** Immediately reduce the price of TV Shows and Video content from $1.99 to $0.99 or $1.49. The data proves that consumers are currently rejecting the $1.99 price point, leaving this media as dead inventory.
2. **Targeted Canadian Campaigns:** Launch targeted marketing campaigns, discounts, or bundles specifically in Canada. Increasing their Average Order Value (AOV) to match the global average represents the easiest path to immediate revenue growth.
3. **Automated Lifecycle Marketing:** Since the average time between purchases is ~132 days, marketing should implement an automated "Recommended for You" email trigger sent at the 120-day mark to reliably capture the customer's next natural purchase cycle.
4. **Artist Bundling:** Market Basket Analysis revealed that specific artists (e.g., Jimi Hendrix) have tracks that are heavily purchased together in the exact same cart. The product team should create "Artist Completion Bundles" offered at a slight discount to drive higher volume checkouts.